
بسم الله الرحمن الرحيم

Cirrhosis is a consequence of chronic liver disease characterized by replacement of liver tissue by fibrosis, scar tissue and regenerative nodules leading to progressive loss of liver function.

The word "cirrhosis" derives from Greek κίρρὸος, meaning *tawny* (the orange-yellow colour of the diseased liver).
While the clinical entity was known before, it was René Laennec who gave it the name "cirrhosis" in his 1819 work.

Causes

Chronic hepatitis C .
Chronic hepatitis B .
Non-alcoholic steatohepatitis (NASH).
Alcoholic liver disease (ALD).
Primary biliary cirrhosis .
Primary sclerosing cholangitis .
Autoimmune hepatitis .
Cardiac cirrhosis .

Hereditary hemochromatosis .
Wilson's disease .
Alpha 1-antitrypsin deficiency .
Galactosemia .
Glycogen storage disease type IV .
Cystic fibrosis .
Hepatotoxic drugs or toxins .
Idiopathic, i.e., of unknown cause .

Signs and symptoms

* *Liver :* Enlarged, normal, or shrunken.
* *Splenomegaly* :Due to congestion of the red pulp as a result of portal hypertension.
* *Ascites* : Accumulation of fluid in the peritoneal cavity. It may be associated with hydrocele.

* *Jaundice* : Yellow discoloring of the skin, eye, and mucus membranes due to increased bilirubin. Urine may also appear dark.
* *Asterixis* : Bilateral asynchronous flapping of outstretched, dorsiflexed hands.

* *Palmar erythema* : Exaggerations of normal speckled mottling of the palm due to altered sex hormone metabolism.
* *Spider nevi* :Vascular lesions consisting of a central arteriole surrounded by many smaller vessels due to an increase in estradiol.

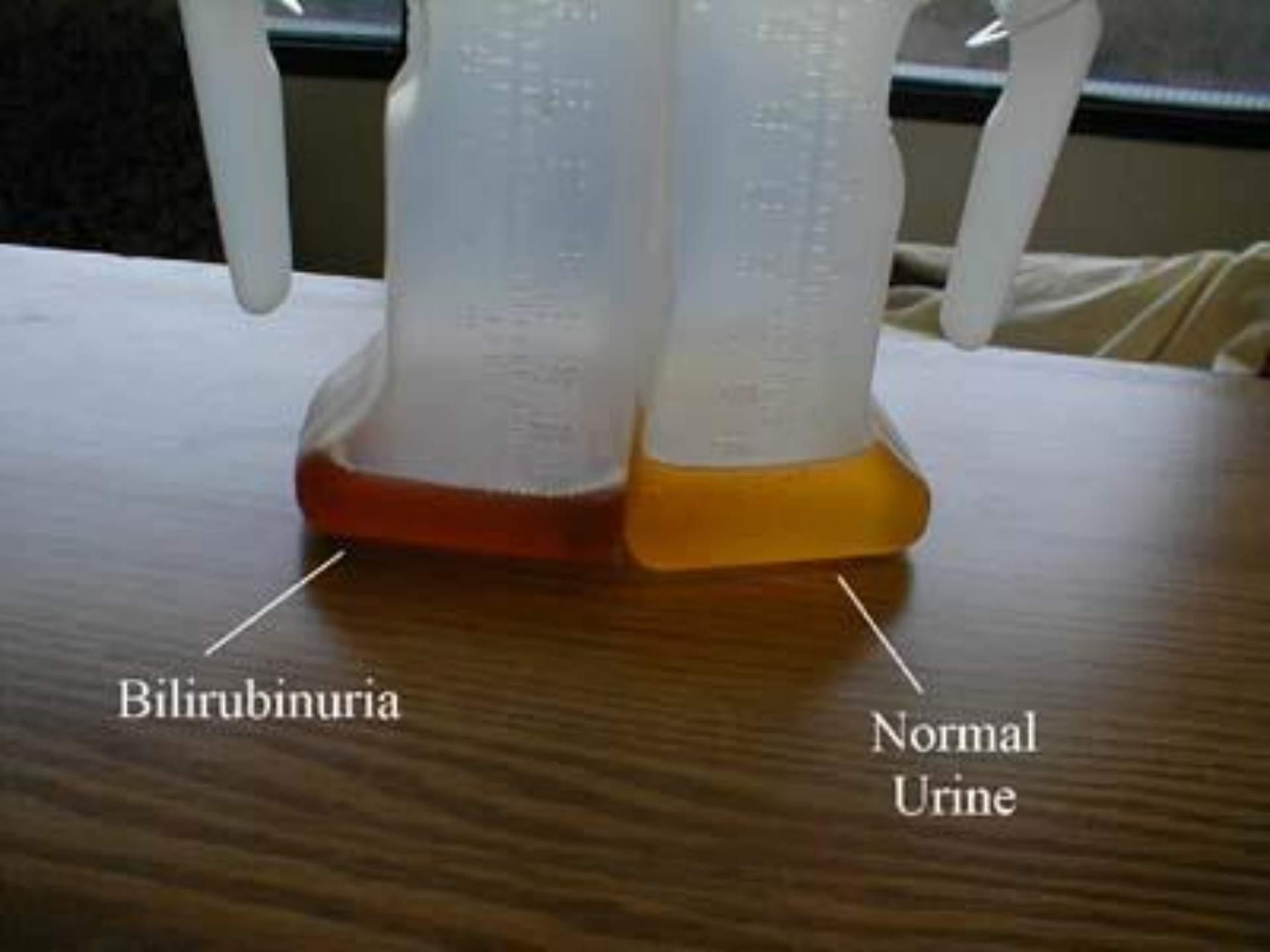
* White *nails* : proximal two-thirds of the nail plate appears white with distal one-third red, due to hypoalbuminemia.
* *Clubbing* : angle between the nail plate and proximal nail fold > 180 degrees.
* *Gynecomastia* : proliferation of male breasts.

- * *Caput medusa* : In portal hypertension, the umbilical vein may open. Blood from the portal venous system may be shunted through the periumbilical veins into the umbilical vein and ultimately to the abdominal wall veins, manifesting as caput medusa.
- * *Fetor hepaticus* : Musty odor in breath due to increased dimethyl sulfide.
- * *Others* : Weakness, fatigue, anorexia, and weight loss.



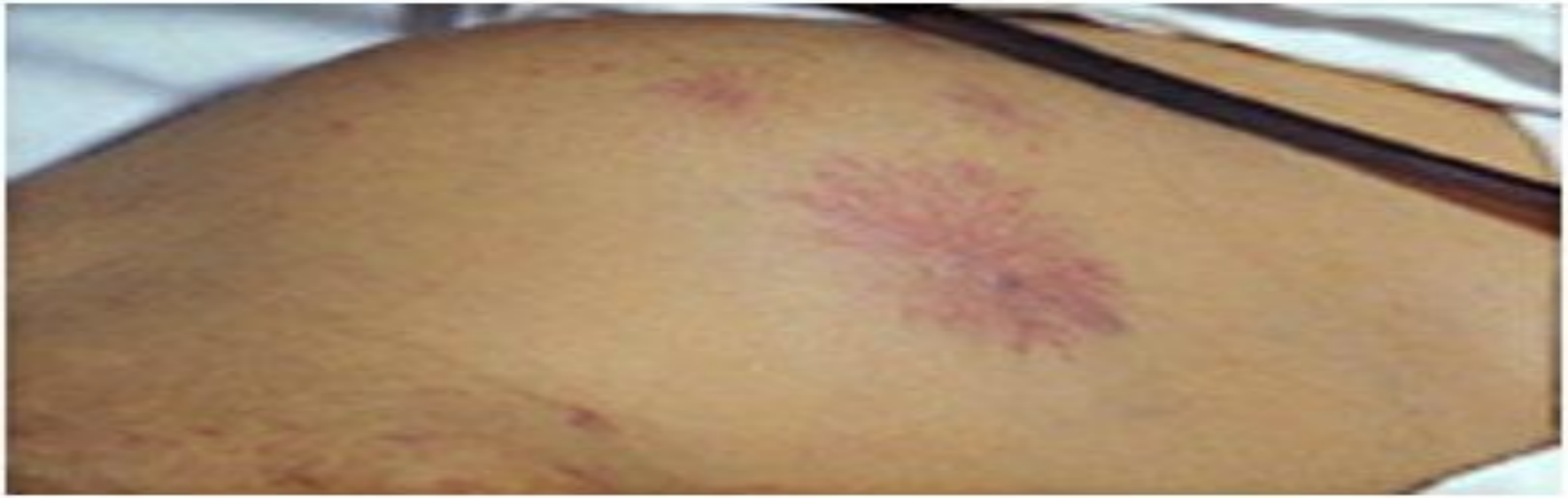


Jaundice


Bilirubinuria
Normal
Urine











Spider naevi










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L.L. edema







Caput medusa


ASCITES



White nails







Complications

Bleeding: decreased production of coagulation factors.
Jaundice: decreased processing of bilirubin.
Itching (pruritus): bile salts deposited in the skin.
Hepatic encephalopathy: The liver does not clear ammonia, carried to the brain, affecting cerebral functioning.

Portal hypertension: leads to the following complications:

Ascites, spontaneous bacterial peritonitis .
Esophageal varices: collateral vessels in the esophagus. These may become enlarged and are more likely to burst.

Hepatocellular carcinoma: It has a high mortality rate.
Sensitivity to medication: due to decreased metabolism of the active compounds.
Hepatorenal syndrome.
Hepatopulmonary syndrome .
Immune system dysfunction leading to infection.

Diagnosis

The gold standard for diagnosis is a liver biopsy.
Histologically cirrhosis can be classified as micronodular, macronodular, or mixed.
However, a biopsy is not necessary if the clinical, laboratory, and radiologic data suggests cirrhosis. Furthermore, there is a significant risk to liver biopsy.

Lab findings:

Aminotransferases: AST and ALT are moderately elevated, with AST > ALT. However, normal aminotransferases do not preclude cirrhosis.

Bilirubin: may elevate as cirrhosis progresses.

Albumin: levels fall.

Prothrombin time: increases due to decreased clotting factors.
Thrombocytopenia: due to both congestive splenomegaly and decreased thrombopoietin from the liver.
Leukopenia and neutropenia: due to splenomegaly with splenic margination.

Imaging:

Ultrasound **is routinely used in the evaluation of cirrhosis.** It also screens for hepatocellular carcinoma and portal hypertension.

Abdominal CT and liver/bile duct MRI (MRCP) in certain circumstances.

Endoscopy:

Endoscopic examination of the esophagus, stomach and duodenum is performed in patients with established cirrhosis to exclude the possibility of esophageal or gastric varices.

Management

Liver damage from cirrhosis cannot be reversed, but treatment could stop or delay further progression and reduce complications.

A healthy diet is encouraged, as cirrhosis may be an energy-consuming process.

Close follow-up is often necessary.

Antibiotics will be prescribed for infections.

Treating underlying causes:

Alcoholic cirrhosis is treated by abstaining from alcohol.
Corticosteroids for autoimmune hepatitis-related cirrhosis.
Cirrhosis caused by Wilson's disease, in which copper builds up in organs, is treated with chelation therapy (e.g., penicillamine) to remove the copper.

Preventing complications:

Ascites:

Salt restriction is often necessary, as cirrhosis leads to accumulation of salt (sodium retention). Diuretics may be necessary to suppress ascites.

Esophageal variceal bleeding:

Sclerotherapy to stop bleeding.

Propranolol is a commonly used agent to lower portal hypertension.

In severe complications, transjugular intrahepatic portosystemic shunting is occasionally indicated.

Hepatic encephalopathy

Spontaneous bacterial peritonitis

Hepatorenal syndrome

Liver Transplantation:

If complications cannot be controlled or when the liver ceases functioning, liver transplantation is necessary. Survival from liver transplantation has been improving and the five-year survival rate is now around 80%.

It depends largely on the severity of disease and other medical problems in the recipient. Transplantation necessitates the use of immune suppressants (cyclosporine or tacrolimus).